William Bekerman

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Department of Statistics and Data Science
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Education

University of Pennsylvania, The Wharton School — Philadelphia, PA
Doctor of Philosophy, Statistics — Expected Dec. 2026
Committee: Dylan Small (advisor), Eric Tchetgen Tchetgen, Zhimei Ren
Dissertation: Data-Driven Designs for Robust Causal Inference in Observational Studies

Cornell University, College of Agriculture and Life Sciences — Ithaca, NY
Bachelor of Science, Biometry & Statistics, *summa cum laude* — 2018–2022
Advisor: Joe Guinness

Research Interests

Causal Inference; Multiple Hypothesis Testing; Exploratory Data Analysis; Applications

Papers and Preprints

[8] **Bekerman, W.**, Mehta, A., Hasson, R.E., Robinson, L.E., Small, D.S., and Fogarty, C.B. (2026). Powerful multivariate sensitivity analysis via sample splitting in an observational study of the effects of poverty on cardiovascular disease risk factors. *Under revision at Journal of the American Statistical Association*. [Preprint]

[7] **Bekerman, W.**, Dalal, A., del Ninno, C., and Small, D.S. (2026). Planning for gold: Hypothesis screening with split samples for valid powerful testing in matched observational studies. *Biometrika*, 113(1), asaf078. [Paper] [Code]

[6] Zhang, R., Kong, J., Small, D.S., and **Bekerman, W.** (2025). Beyond the ACE score: Replicable combinations of adverse childhood experiences that worsen depression risk. [Preprint] [Study protocol]

[5] Jin, H.A., **Bekerman, W.**, Small, D.S., and Rabinowitz, A. (2024). An observational study on effects of contact, collision, and non-contact sports participation on cognitive and emotional health. *Under revision at PLOS One*. [Study protocol]

[4] Mi, X., **Bekerman, W.**, Sims, P.A., Canoll, P.D., and Hu, J. (2024). RZiMM-scRNA: A regularized zero-inflated mixture model framework for single-cell RNA-seq data. *Annals of Applied Statistics*, 18(1), 1–22. [Paper] [Code]

[3] **Bekerman, W.**, Bogomolov, M., Heller, R., and Small, D.S. (2024). Protocol for an observational study on the effects of paternal alcohol use disorder on children's later life outcomes. [Preprint]

[2] **Bekerman, W.**, and Guinness, J. (2023). Comparison of CYGNSS and Jason-3 wind speed measurements via Gaussian processes. *Data Science in Science*, 2(1), 2194349. [Paper] [Code]

[1] **Bekerman, W.**, and Srivastava, M. (2021). Determining decomposition levels for wavelet denoising using sparsity plot. *IEEE Access*, 9, 110582–110591. [Paper] [Code]

Awards and Honors

Lawrence D. Brown Student Paper Award	2026
IMS Hannan Graduate Student Travel Award	2025
NSF Graduate Research Fellowship	2022
Merrill Presidential Scholar (*top 1% of graduating class*)	2022
Hunter R. Rawlings III Cornell Presidential Research Scholar	2020
First Prize, Cornell Bits on Our Minds Showcase	2019

Presentations

Lawrence D. Brown Student Paper Award Symposium	Nov. 2026
American Causal Inference Conference	May 2026
European Causal Inference Meeting	Apr. 2026
Online Causal Inference Seminar	Mar. 2026
UPenn Center for Causal Inference, UPenn	Sep. 2025
International Conference on Multiple Comparison Procedures	Aug. 2025
Joint Statistical Meetings	Aug. 2025
American Causal Inference Conference	May 2025
ENAR Spring Meeting, New Orleans	Mar. 2025
UPenn Selective Inference Reading Group	Feb. 2025
International Seminar on Selective Inference	Oct. 2024
UPenn Selective Inference Reading Group	Oct. 2024
UPenn Center for Causal Inference	Sep. 2024
UPenn Conference on Big Data in Biomedical & Population Health Sciences	Sep. 2024
American Causal Inference Conference	May 2024
Biophysical Society Annual Meeting	Feb. 2022
Joint Statistical Meetings	Aug. 2021

Teaching

Teaching Assistant, Cornell University

BTRY 4030: Linear Models with Matrices	Fall 2021
BTRY 6020: Statistical Methods II (*graduate-level*)	Spring 2021
INFO 2950: Introduction to Data Science	Fall 2020
PLBIO 2400: Green World, Blue Planet	Fall 2019, 2020, 2021

Mentoring

Undergraduate Research Supervision

Ruizhe Zhang (2024–2026); *Current:* PhD student, Statistics, Stanford University
Matthew Spivey (2024–2025); *Current:* MS student, Mathematics, Penn
Hannah Jin (2022–2024); *Current:* PhD student, Biostatistics, Harvard University

Other. Peer mentor for Penn Statistics PhDs, Cornell Biometry & Statistics undergraduates

Service

Academic Leadership

Co-organizer, Wharton Statistics Student Seminar
Inaugural Student Representative, Society for Causal Inference (SCI)
Chair, SCI Student & Postdoc Committee

Journal Reviewing

IEEE Transactions on Information Theory
Journal of Biopharmaceutical Statistics
Journal of Quantitative Analysis in Sports
Journal of the Royal Statistical Society – Series B (x2)

Judging

Long Island Science and Engineering Fair (*intelligent machines, math, software, robotics*)

Other

Alpha Phi Omega Service Fraternity (*leadership/friendship/service award recipient*)
Cornell Data Science Project Team (*team lead, project lead, recruitment chair*)

Industry Experience

Pharmaceutical Consulting – ZS Associates 2021
Decision Analytics Intern

- Optimized the efficiency of an omni-strategy messaging campaign for AstraZeneca.
- Identified statistically significant engagement and retention discrepancies due to campaign overlap and messaging frequency.
- Presented detailed, action-oriented recommendations during client meetings.